



Bilkent University
Department of Computer Engineering

Senior Design Project
T2333
JobTalk

Design Project Final Report

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1. Introduction

Recruitment is essential for company success since it impacts talent quality. Rising candidate numbers and diverse qualifications pose challenges in recruitment. Online methods streamline the process, yet evaluating data remains complex. HR may lack technical expertise, necessitating an unbiased approach for effective hiring.

JobTalk aims to improve the efficiency of the recruitment process by saving resources and making impartial evaluations. Upon recruiters' request for interviews, candidates participate in conference calls where they respond to questions, which are then analyzed by the system. JobTalk's algorithms assess candidates based on sentiment analysis, self-expression, communication skills, and the content of their answers, aligning with the recruiter's desired criteria. The evaluations are synthesized into a comprehensive report for recruiters to review.

JobTalk eliminates human bias and ensures a fair evaluation process by using machine learning algorithms that are not subject to judgment based on appearance, ethnicity or background. Furthermore, JobTalk speeds up the recruitment process, requiring less action from the human resources department. By providing an automated report on the candidate, JobTalk improves the overall recruitment performance.

2. Requirements Details

JobTalk is an advanced recruitment platform that leverages machine learning to analyze candidate interviews, streamlining the hiring process and removing bias. JobTalk comprises two main client types: companies and candidates. Companies register on JobTalk by subscribing to the service, which provides them with company administrator and recruiter accounts to manage the hiring process. Companies use JobTalk to improve recruitment efficiency, reduce bias, and enhance the quality of talent selected.

JobTalk's machine learning algorithms analyze candidates' responses using various metrics, including sentiment analysis, self-expression, communication skills, and the content of their answers. These evaluations are compiled into a comprehensive report, which is then presented to the recruiters for review. The system eliminates human bias by using algorithms not influenced by appearance, ethnicity, or background.

Below are the functional requirements based on the user types:

All users:

- There exist four types of users: system admin, company admin, recruiter, and candidate.
- All users must register/log in using their emails.

System admins:

- System admins can register company admins to the system where the registered user gets an email with their password for the first login.
- System admins can request interviews from candidates which involves registration of that user where they get an email with their password for the first login.

Company admins:

- Company admins can create/delete/update their companies.
- Company admins can create/delete/update recruiters.
- Company admins can create/delete/update interview guides.
- Company admins can create/delete/update jobs.
- Company admins can create/delete/update questions on interview guides.
- Company admins can change the “hiring status” of jobs to “hiring”/“not hiring”.

Recruiters:

- Recruiters can see the list of the interviews in two categories: past and ongoing.
- Recruiters can request an interview from a candidate specifying the guide code, job code, candidate’s email address, and the due date.
- Recruiters can create/edit/delete interview guides.
- Recruiters can create/delete/update jobs.
- Recruiters can create/delete/update questions on interview guides.
- Recruiters can change the “hiring status” of jobs to “hiring”/“not hiring”.
- Recruiters can see the detailed analysis of each interview produced by the ML algorithms.

Candidates:

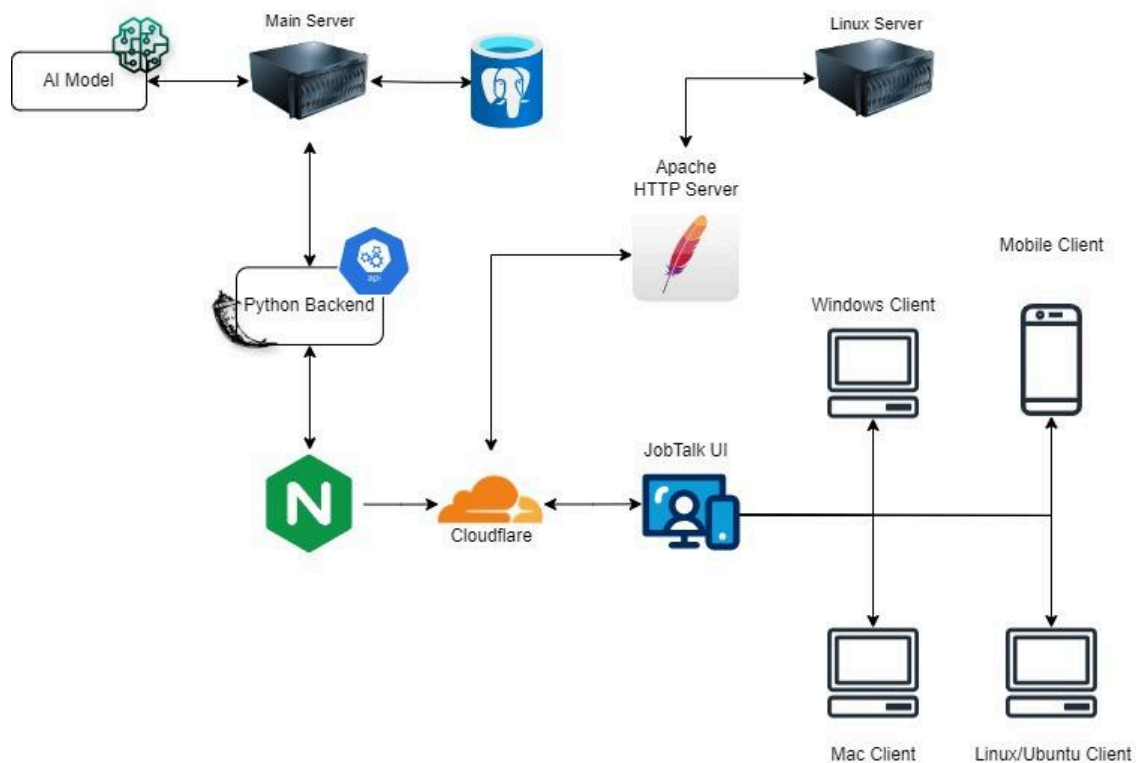
- Candidates can see the list of their current/past interviews.
- Candidates can join an interview through the “join” button next to the related interview.

- The questions will be shown one by one, expecting an answer for each question in the given period.
- Candidates can start/stop recording for each question.
- Candidates can submit their interview once all questions are answered.

System:

- The system should analyze the recordings for effective speech characteristics, content, and to generate a report about the interview.
- The system should train and use a machine learning model for emotion recognition, which can classify voice samples into predefined emotional categories (anger, disgust, neutral, fear, happy, sad, stress, confidence, etc.).
- The system should transcribe the audio and prompt a large language model for content analysis and score the candidate based on the questions and keywords.
- Analysis reports will be generated according to the emotional analysis and content analysis.
- The system should allow each recruiter to be associated with only one company.

3. Final Architecture and Design Details



The JobTalk system integrates various components and servers to provide a comprehensive recruitment solution that uses AI algorithms to evaluate candidates. The system is designed to support multiple client types, including Windows, macOS, Linux/Ubuntu, and mobile platforms.

AI Model: The core of the system is an AI model that processes and evaluates candidate data. This model is central to the recruitment tool's functionality.

Main Server: A robust main server acts as the central processing unit for the system. It manages data flow between the AI model, and the Python backend.

Python Backend: The backend logic is handled by a Python-based application, which interacts with the AI model and the frontend UI through a series of APIs.

Nginx: Nginx is configured as a reverse proxy to efficiently manage requests between the Internet (via Cloudflare) and the internal network that hosts the Python backend and other services.

Cloudflare: Cloudflare is used to secure and accelerate the Internet connection to the Python backend. It provides DDoS protection and traffic optimization. Cloudflare's WAF is utilized for security.

Apache HTTP Server: Hosted on a Linux server, the Apache HTTP Server handles the delivery of the JobTalk UI to various clients. It ensures that the user interface is accessible across different platforms.

JobTalk UI: The user interface is accessible to various clients, including Windows, macOS, Linux/Ubuntu, and mobile devices. It provides a consistent and responsive experience across all platforms.

All components are interconnected using secure and efficient protocols to ensure data integrity and system performance. The main server acts as the hub for internal data flow, while Cloudflare and Nginx manage external access to the system.

The system incorporates advanced security measures through the use of Cloudflare for threat management and SSL/TLS protocols for secure data transmission.

This architecture supports robust, scalable, and secure operations for the JobTalk recruitment system, leveraging advanced technologies to optimize candidate evaluation and user experience.

4. Development/Implementation Details

4.1. Analysis Pipeline

To develop an automated workflow for analyzing and predicting scores for job interview recordings, we have divided the process into two main components: content analysis and performance analysis. **Content Analysis:** This segment focuses on scrutinizing the interview transcript to determine if the candidate's responses align with the job profile specified by the company. It involves a detailed examination of the content provided by the candidate to assess its relevance and appropriateness. **Performance Analysis:** This aspect evaluates how the interviewee delivered their responses during the interview. Key factors considered include the emotion and engagement demonstrated, speaking ability, and the specific word choices used to articulate responses. This analysis helps gauge the candidate's communicative effectiveness and overall presentation skills.

Figure 2, illustrates the high-level flow of this automated scoring workflow, clearly delineating the steps involved in both content and performance evaluations.

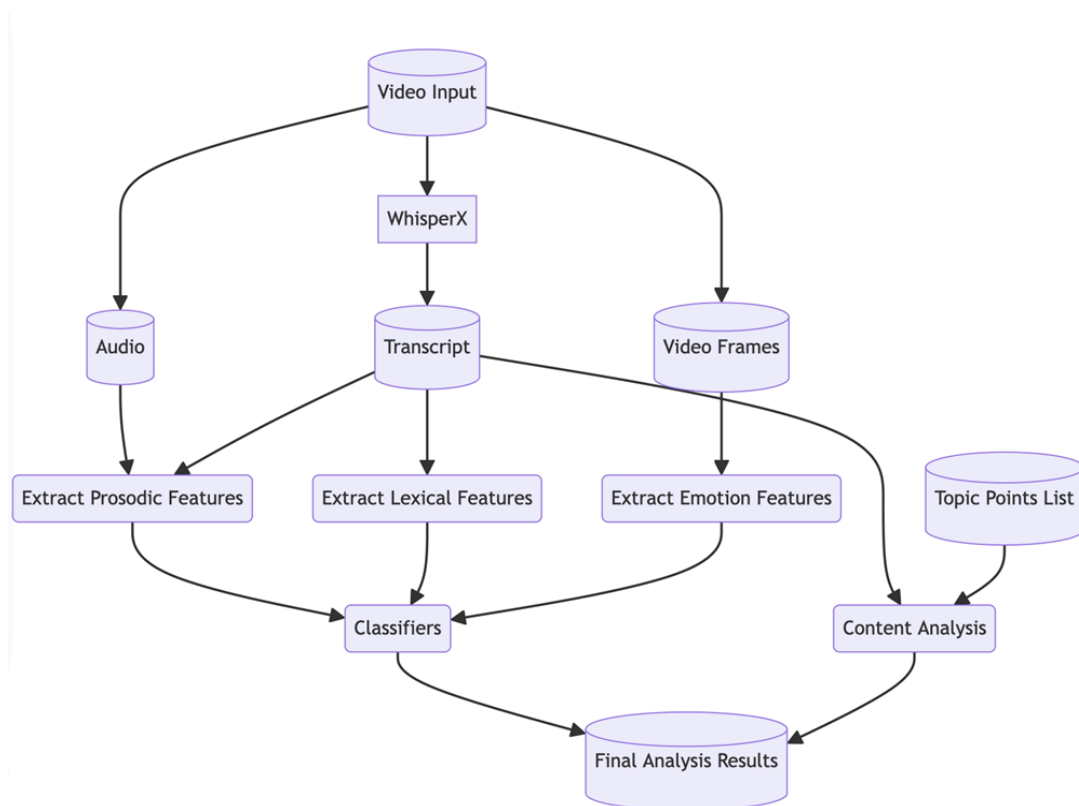


Fig. 2: High level flow

4.1.1. Speech Performance Analysis

For assessing candidate’s speech performance, we utilized the MIT job interview dataset [6] to train multiple models that capture various aspects of interview dynamics. N. Iftekhhar, T. Glidea, D. Mohammed, and H. Ehsan suggest a multimodal machine learning pipeline to analyze these interviews [5].

Feature extraction for model training is categorized into three groups. The first set of features relates to speech delivery. These prosodic features include rhyme, stress, intonation, speaking rate, and other speech dynamics. The paper [5] recommends using Praat, an open-source software capable of analyzing low-level prosodic features like frequency content, jitters, shimmers, and speech energy. We expanded on this foundation to include additional mid-level features such as silent and filled pauses (e.g., "uhhms," "umms"), speech rate, and word chunk lengths. The specifics of prosodic feature extraction are detailed in section 1.1, "Prosody Analysis."

The second class of features pertains to the candidate's facial expressions. We analyzed several emotions extracted from each video frame, with the methodologies and findings documented in section 1.2, "Emotional Analysis."

Lastly, the third feature class focuses on lexical choices, examining the distribution and usage of words within the speech. This aspect is crucial for assessing the candidate's language skills and alignment with job requirements. Further details on this are available in section 1.3, "Lexical Analysis."

4.1.1.1. Prosody Analysis

We computed the following sets of low-level features using the Praat software.

Feature Category	Feature Name	Description
Formant Features	f1_mean	Mean frequency of the first formant (F1)
	f2_mean	Mean frequency of the second formant (F2)
	f3_mean	Mean frequency of the third formant (F3)
	f4_mean	Mean frequency of the fourth formant (F4)
	f1_median	Median frequency of the first formant (F1)

	f2_median	Median frequency of the second formant (F2)
	f3_median	Median frequency of the third formant (F3)
	f4_median	Median frequency of the fourth formant (F4)
Pitch Features	duration	Total duration of the audio recording
	meanF0	Mean fundamental frequency (F0)
	stdevF0	Standard deviation of the fundamental frequency (F0)
	hnr	Harmonics-to-noise ratio
	local_jitter	Measure of frequency variation
	local_absolute_jitter	Absolute local jitter
	rap_jitter	Relative average perturbation jitter
	ppq5_jitter	Five-point period perturbation quotient jitter
	ddp_jitter	Difference of differences of periods jitter
	local_shimmer	Local shimmer measure
	localdb_shimmer	Shimmer measured in decibels
	apq3_shimmer	Three-point amplitude perturbation quotient shimmer
	apq5_shimmer	Five-point amplitude perturbation quotient shimmer
	apq11_shimmer	Eleven-point amplitude perturbation quotient shimmer
	dda_shimmer	Difference of differences of amplitudes shimmer
Energy Features	mean_spectral_energy	Mean of the spectral energy of the recording
	intensity_mean	Mean intensity of the recording
	intensity_min	Minimum intensity of the recording
	intensity_max	Maximum intensity of the recording
	intensity_range	Range of intensity in the recording

	intensity_sd	Standard deviation of intensity in the recording
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Table 1: Low-level features

We wanted to go beyond these low-level features since using filler words, pauses, and speech rate greatly captures a speaker's speaking ability and characteristics.

A system called Speech Rater [1][2][3] is proposed to score spoken IELTS test responses automatically. Speech Rater utilizes a custom-made ASR to make word-level timestamped transcriptions on which they calculate the features. A critical aspect of their custom ASR is capturing important structural events such as filled pauses and disfluencies in the speech.

This is a significant shortcoming of many off-the-shelf ASR tools since they are internally designed to filter out such sounds. Therefore, a VAD is employed to detect filled pauses in the speech, as described in[4].

Our pipeline starts with a word-level timestamped transcription. We decided to use Whisper [7] since it is extremely powerful and open-source. However, Whisper can only provide segment-level timestamps. Segments are similar structures to sentences. To tackle the problem of word-level timestamps, we used a variation of whisper named whisper-x[8], which does forced alignment with the transcription and the audio file. Using whisper-x, we obtained a word-level timestamped transcription. From this transcription, we gathered the features in Table 2.

Feature Category	Feature Name	Description
General Features	total_duration	Total duration of the audio recording in seconds.
	total_words	Total number of words in the transcription of the audio.
	unique_words_count	Number of unique words in the transcription.
Text Analysis	average_chunk_length_in_words	Average number of words per segment/chunk in the transcription.
	mean_deviation_of_chunks_in_words	Average deviation in the number of words per chunk, indicating variability.

Speech Dynamics	articulation_rate	Number of words spoken per minute, indicating the speed of speech.
	duration_of_silences_per_word	Average duration of silences normalized by the number of words.
	mean_of_silence_duration	Average duration of all silence intervals in the recording.
	mean_duration_of_longer pauses	Average duration of longer pauses within the speech.
	frequency_of_longer pauses_divided_by_number_of_words	Frequency of longer pauses relative to the total number of words.
Linguistic Complexity	types_divided_by_uttdur	Ratio of unique word types to the total duration of speech segments.
Pause Analysis	mean_length_of_filled pauses	Average length of filled pauses (e.g., "uh", "um") in the speech.
	frequency_of_filled pauses	Frequency of occurrence of filled pauses in the speech.
Assessment	asr_score	An overall score or rating derived from automatic speech recognition (ASR) analysis.

Table 2: Features obtained from the transcription

We defined a word chunk as a sequence of words with a gap between them under a fixed threshold duration. After identifying the chunks, we calculated the Text Analysis features. Identifying filled pauses was another major problem we had to tackle since Whisper treats those filled pauses as silences and filters them so that they are not in the final transcript. We utilized the method described by G. Zhu, J. Caceres, and J. Salamon [4]. The algorithm first identifies the regions where the ASR is not triggered. These regions are either silences or filled pauses without a spoken word. We then run a VAD over these identified regions to identify which parts contain a voice and which are silent. The algorithm described in [4] further passes the voiced regions to a classifier model to detect if the part is a filled pause like umm, hmm, or another type of voiced part such as laughter or breaths. We implemented the classification by thresholding the length of the voiced parts. Shorter regions tend to have short bursts of breaths, and more significant regions tend to be filled with pauses.

4.1.1.2. Lexical Analysis

In this class of features, we calculate a histogram of certain word classes from the transcript. Table 3 describes the histogram and the content of bins.

LIWC Category	Description	Examples
Pronouns	I	I, I'm, I've, I'll, I'd, etc.
	We	we, we'll, we're, us, our, etc.
	They	they, they're, they'll, them, etc.
Speech Disfluencies	Non-fluencies	Words introducing non-fluency in speech, e.g., uh, umm, well.
Emotions	PosEmotion	Words expressing positive emotions, e.g., hope, improve, kind, love.
	NegEmotion	Words expressing negative emotions, e.g., bad, fool, hate, lose.
	Anxiety	Nervous, obsessed, panic, shy, etc.
	Anger	Agitate, bother, confront, disgust, etc.
	Sadness	Fail, grief, hurt, inferior, etc.
Cognitive	Cognitive	Cause, know, learn, make, notice, etc.
	Inhibition	Refrain, prohibit, prevent, stop, etc.
	Perceptual	Observe, experience, view, watch, etc.
	Relativity	First, huge, new, etc.
Work-Related	Work	Project, study, thesis, university, etc.
Informal Speech	Swear	Informal and swear words.
Grammar	Articles	A, an, the, etc.

	Verbs	Common English verbs.
	Adverbs	Common English adverbs.
	Prepositions	Common prepositions.
	Conjunctions	Common conjunctions.
	Negations	No, never, none, cannot, don't, etc.
	Quantifiers	All, best, bunch, few, ton, unique, etc.
	Numbers	Words related to number, e.g., first, second, hundred, etc.

Table 3: Word Histogram

4.1.1.3. Emotional Analysis

In order to capture the emotional state of the interviewee, we calculated five features from the video.

Variable Name	Description
angry	Indicates the level of anger.
disgust	Indicates the level of disgust.
fear	Indicates the level of fear.
happy	Indicates the level of happiness.
neutral	Indicates a neutral emotional state.

Table 4: Emotion-Related Features

We used a Python package called *fer* to obtain these features for each frame. For a frame, *fer* gives a probability distribution over the emotion classes. After processing the video, we averaged the values along the time axis. That is, we took the average of all frames. Paper[5] describes a similar method where they have a binary classifier that can identify between smiling and non-smiling faces.

4.1.1.4. Regression Model Training & Results

After computing all the features mentioned in 1.1,1.2,1.3, we concatenated them to a single vector. We then normalize each feature to mean 0 stdev 1. In the MIT dataset [6], each interview is scored under 18 categories. We trained a different SVR(Support Vector Regression) for each scoring category. AUC indicates how well the model can sort the inputs for a regression model. In the dataset, we had the following 18 scores for interviews: Overall, RecommendHiring, Colleague, Engaged, Excited, EyeContact, Smiled, SpeakingRate, NoFillers, Friendly, Paused, EngagingTone, StructuredAnswers, Calm, NotStressed, Focused, Authentic, NotAwkward. We chose the models with high correlation and included them in our results and analysis pipeline.

We used 20% of the data to test our models. To remove any bias that can result in selecting a specific subset of the data for testing. We did 1000 different test splits and had the mean of the correlation and AUC scores.

Model Name	Correlation	AUC
RecommendHiring	0.6122	0.7727
Colleague	0.5484	0.7662
Engaged	0.6468	0.7894
Excited	0.7463	0.8716
Smiled	0.6950	0.8321
Friendly	0.7076	0.8144
Paused	0.5266	0.7246
EngagingTone	0.7355	0.8538
StructuredAnswers	0.6191	0.7896

Table 5: Trained Model Statistics

4.1.2. Content Analysis

This section outlines a flexible content analysis process utilizing a Large Language Model (LLM), specifically employing OpenAI's GPT-4 API. Our approach begins by identifying a set of expected topics within the interviewee's responses. Unlike traditional methods that rely on searching transcripts for specific keywords, using an LLM enhances our flexibility in two significant ways:

Firstly, our analysis can interpret both sentences and questions as topic points, in addition to accepting single words. For example, rather than merely searching for a keyword, we can analyze complex statements such as "The interviewee provided examples of their previous teamwork experiences," or "The interviewee prefers working independently."

Secondly, the LLM allows for contextual interpretation of keywords. For instance, if the topic point is "Relational Database," and the transcript mentions "SQL," the LLM can infer the presence of content related to relational databases, even if the exact phrase isn't used. This capability significantly broadens the scope of our analysis, ensuring more nuanced content assessment.

The evaluation of each topic point is quantified as follows: the LLM assigns a score of 0 if the topic is absent from the response, 1 if the topic is partially or weakly addressed, and 2 if the topic is fully covered. After assessing each point, the LLM provides a summary that includes whether and how each topic is addressed or mentioned.

Finally, we calculate a content score on a scale of 1. This score is determined by the formula:

$$\text{Content Score} = \text{Sum of topic scores} / \text{number of topics} * 2$$

This method allows us to quantitatively measure the relevance and completeness of the interviewee's answers in relation to the expected topics, providing a clear metric to gauge content alignment.

4.2. Data Processing Pipeline

4.2.1. Recording Technology

The front-end part of the application uses the MediaStream Recording API to create a WebM stream with Opus codec that is specifically used in WebRTC applications. This WebM stream is captured when the "Record" button is pressed, encoded with base64, and sent to the back-end over the websocket connection that is created by using "Socket.IO".

On the back-end side, Flask-SocketIO captures the packet, decodes it into bytes, and writes it onto a WebM file. After the recording stopped, these WebM files are converted into MP4 and WAV files to be used in our interview analysis. This whole procedure is repeated for the recording of every question. Before the transmission of the media stream, necessary information about candidates, interviews, and questions are sent over a websocket that enables the application to sort and classify each recording with proper naming and branching on the file system.

4.2.2. Background Service

The analysis module used in our system is resource intensive, since it handles both video and audio processing. Given the complexity and demand on system resources, processing time is substantial. To solve this problem, we have designed and implemented a background service. This service continuously interacts with the database to select interviews marked for analysis.

When interviews are collected from the database, they are fed into the analysis module one by one. This module processes each recording, extracting necessary data and feeding the data into the machine learning models. After the analysis is completed and saved, the status of the interview is updated to indicate that it has been processed to make sure that an interview is processed only once, saving system resources.

The service is developed using Python, which offers libraries compatible with our web application. For the service to run continuously, it is managed via systemd, a system and service manager for Linux operating systems, in Ubuntu.

5. Test Cases and Results

5.1. Functional Test Cases

Test ID	Test 1	Test Type/ Category	Integration Testing	Severity	Critical
Objective	Verify user can login				
Test Steps	● Launch the application ● Open the application login page ● Input valid username and password in the respective fields ● Click on the "Login" button ● Wait for the system to authenticate the provided credentials ● Verify if the user is successfully logged in and directed to the				

	dashboard or home page
Expected Results	• Upon successful login, the user should be redirected to the application's dashboard. • The user's name should be displayed to indicate that they are logged in.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 2	Test Type/Category	Component Testing	Severity	Major
Objective	Verify “add company” button works				
Test Steps	• Log in as a company admin • Identify and locate the "Add Company" button on the user interface • Click on the "Company Add" button • Check if the company add form is shown on the page				
Expected Results	• Upon clicking the "Company Add" button, a form for adding a new company should appear. • All required fields for adding a company should be clearly labeled and accessible.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 3	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify company can be added				
Test Steps	• Identify and locate the "Company Add" button on the user interface • Click on the "Company Add" button • Check if the company add form is shown on the page • Fill in the required information for adding a new company, such as company name, address, contact details, etc. • Submit the form or save the company details by clicking on add company button. • Verify if a success message is displayed confirming that the company has been successfully added. • Check if the newly added company appears in the list of				

	companies displayed on the page.
Expected Results	• After clicking on the "Add Company" button, a form or dialog for adding a new company should be displayed. • All required fields for adding a company should be clearly labeled and accessible. • Upon entering the company details and submitting the form, a success message should be displayed indicating that the company has been added successfully. • The newly added company should be visible in the list of companies on the page. • The added company's details should be accurately reflected in the list, including the company name, address, and other relevant information.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 4	Test Type/Category	Component Testing	Severity	Major
Objective	Verify company admin add button works				
Test Steps	• Launch the application. • Locate the "Company Admin Add" button. • Click on the "Company Admin Add" button.				
Expected Results	• A form or dialog for adding a new company admin should appear after clicking the "Company Admin Add" button. • All mandatory fields for adding an admin should be accessible and clearly labeled.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 5	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify company admin is created				
Test Steps	• Launch the application. • Locate the "Company Admin Add" button. • Click on the "Company Admin Add" button. • Enter company admin details. • Submit the form. • Verify company admin can log in with correct credentials.				

Expected Results	• A form for adding a new company admin should appear after clicking the "Company Admin Add" button. • All mandatory fields for adding a company admin should be accessible and clearly labeled. • After submitting the form, the system should process the request without errors. • Upon successful addition, the newly added admin should be displayed in the list of company admins or a success message should be shown. • Validation errors for incomplete or incorrect inputs should be displayed to guide the user. • Newly added company admin should be able to log in with correct credentials.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 6	Test Type/Category	Component Testing	Severity	Major
Objective	Verify create recruiter button works				
Test Steps	• Open the application. • Navigate to company admin homepage. • Identify the "Create Recruiter" button. • Click on the "Create Recruiter" button. • Fill in the required details for creating a new recruiter.				
Expected Results	• Upon clicking the "Create Recruiter" button, a form or dialog for creating a new recruiter should be displayed. • All mandatory fields for creating a recruiter should be clearly labeled and accessible.				
Date Tested and Test Result	08.05.2024 - bug				

Test ID	Test 7	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify recruiter is created				
Test Steps	• Open the application. • Navigate to company admin homepage. • Identify the "Create Recruiter" button.				

	● Click on the "Create Recruiter" button. ● Fill in the required details for creating a new recruiter. ● Submit the form by clicking on the "Save" or "Submit" button. ● Check if the newly created recruiter appears in the list of recruiters. ● Check if the newly created recruiter can log in successfully.
Expected Results	● Upon clicking the "Create Recruiter" button, a form or dialog for creating a new recruiter should be displayed. ● All mandatory fields for creating a recruiter should be clearly labeled and accessible. ● After submitting the form, the system should process the request without errors. ● The newly created recruiter should be visible in the list of recruiters. ● Any validation errors for incomplete or incorrect inputs should be displayed to guide the user. ● Newly added recruiter should be able to log in with correct credentials.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 8	Test Type/ Category	Component Testing	Severity	Major
Objective	Verify add candidate button works				
Test Steps	● Login as a recruiter. ● Locate the "Request Interview" button. ● Click on the "Request Interview" button. ● Fill in the necessary details for adding a new candidate. ● Submit the form by clicking on the "Submit" button. ● Verify if the newly added candidate appears in the list of candidates. ● Verify if the newly added candidate can login				
Expected Results	● After clicking the "Add Candidate" button, a form for adding a new candidate should be displayed. ● All mandatory fields required for adding a candidate should be clearly labeled and accessible. ● Upon submitting the form, the system should process the request without errors. ● The newly added candidate should be visible in the list of candidates.				

	● If there are any validation errors for incomplete or incorrect inputs, appropriate messages should be displayed to guide the user. ● Newly added candidate should be able to log in with correct credentials.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 9	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify candidate is added				
Test Steps	● Access the application. ● Navigate to the section for managing candidates. ● Locate the "Add Candidate" button. ● Click on the "Add Candidate" button. ● Fill in the necessary details for adding a new candidate. ● Submit the form by clicking on the "Submit" button. ● Verify if the newly added candidate appears in the list of candidates.				
Expected Results	● After clicking the "Add Candidate" button, a form for adding a new candidate should be displayed. ● All mandatory fields required for adding a candidate should be clearly labeled and accessible. ● Upon submitting the form, the system should process the request without errors. ● The newly added candidate should be visible in the list of candidates. ● If there are any validation errors for incomplete or incorrect inputs, appropriate messages should be displayed to guide the user.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 10	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify recruiter can send interviews to candidates				

Test Steps	● Log in to the application as a recruiter. ● Navigate to homepage. ● Select a candidate from the list. ● Click on the option to schedule an interview for the selected candidate. ● Fill in the necessary details for scheduling the interview, including date, time, and interview type. ● Confirm the interview scheduling by clicking on the "Send Interview" or "Schedule Interview" button. ● Verify if the interview confirmation is displayed to the recruiter. ● Optionally, check if the scheduled interview details are updated in the candidate's profile.
Expected Results	● The recruiter should be able to select a candidate and initiate the interview scheduling process without encountering any errors. ● All necessary fields for scheduling the interview should be clearly labeled and accessible. ● Upon confirming the interview scheduling, a confirmation message should be displayed to the recruiter. ● The scheduled interview details should be accurately updated in the candidate's profile. ● If any errors occur during the scheduling process (e.g., conflicting schedule, invalid input), appropriate error messages should be displayed to guide the recruiter. ● Successfully login as the candidate and check home page for the sent interview.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 11	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify recruiter can play interview video				
Test Steps	● Log in to the application as a recruiter. ● Navigate to the homepage. ● Select an interview from the list. ● Click on the option to play the interview video. ● Observe the video playback controls for pause, play, seek, and volume. ● Verify if the interview video plays smoothly without buffering or interruptions. ● Optionally, check if additional features like fullscreen mode or playback speed adjustment are available.				

Expected Results	● The recruiter should be able to select an interview and initiate the playback of the interview video without encountering any errors. ● The video playback controls should be functional and responsive, allowing the recruiter to control playback as desired. ● The interview video should play smoothly and without buffering or interruptions. ● Additional features, if available, should function as expected and enhance the user experience. ● If any errors occur during video playback, appropriate error messages should be displayed to guide the recruiter.
Date Tested and Test Result	08.05.2024 - NA

Test ID	Test 12	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify recruiter can view interview analysis				
Test Steps	● Log in to the application as a recruiter. ● Navigate to the section for managing interviews or candidates. ● Select an interview from the list. ● Click on the option to view interview analysis or insights. ● Verify if the interview analysis is displayed, including relevant metrics such as sentiment score, content score and etc. ● Review the analysis details and ensure they provide valuable insights into the interview performance.				
Expected Results	● The recruiter should be able to select an interview and view the analysis without encountering any errors. ● The interview analysis should include relevant metrics and insights that provide valuable information about the interview performance. ● The analysis details should be clear and understandable, allowing the recruiter to interpret the results effectively. ● Additional features, if available, should enhance the usability and usefulness of the analysis. ● If any errors occur during analysis viewing, appropriate error messages should be displayed to guide the recruiter.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 13	Test Type/ Category	Integration Testing	Severity	Minor
Objective	Verify profile is reachable from homepage				
Test Steps	● Log in to the application. ● Navigate to the homepage of the application. ● Look for the user's profile information displayed on the homepage. ● Click on the user's profile information or avatar. ● Verify if the user is redirected to their profile page.				
Expected Results	● The user's profile information or avatar should be visible on the homepage. ● Clicking on the user's profile information or avatar should lead to their profile page without encountering any errors. ● The profile page should display relevant information about the user, such as name, email, role, etc. ● The user should be able to access their profile information from the profile page.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 14	Test Type/ Category	Integration Testing	Severity	Minor
Objective	Verify the user can edit their profiles				
Test Steps	● Log in to the application. ● Navigate to the user's profile page. ● Modify one or more fields in the profile form (e.g., name, email, social media accounts). ● Click on the "Submit" button. ● Verify if the changes are saved successfully. ● Optionally, verify if the updated profile information reflects correctly on the user's profile page.				
Expected Results	● The user should be able to access their profile editing options without encountering any errors. ● The profile editing form should allow the user to modify their profile information, including name, email, password, etc. ● After submitting the changes, the system should process the				

	request without errors, and the changes should be saved successfully. • If any errors occur during the editing process (e.g., validation errors, server errors), appropriate error messages should be displayed to guide the user.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 15	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify the user can change their password				
Test Steps	• Log in to the application. • Navigate to the user's profile page. • Fill in the current password and the new password in the respective fields. • Confirm the new password by re-entering it in the designated field. • Submit the password change request by clicking on the "Save" button. • Verify if the password change is successful by logging out and logging back in with the new password.				
Expected Results	• The user should be able to access the password change options without encountering any errors. • The password change form should require the user to provide their current password for security reasons. • After submitting the password change request, the system should process the request without errors, and the password should be changed successfully. • Upon successful password change, the user should be able to log in using the new password. • If any errors occur during the password change process (e.g., validation errors, server errors), appropriate error messages should be displayed to guide the user.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 16	Test Type/Category	Integration Testing	Severity	Major
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Objective	Verify the user can logout
Test Steps	● Log in to the application. ● Click on the "Logout" button on the navbar. ● Verify if the user is successfully logged out. ● Optionally, try accessing any restricted pages or functionalities to confirm that the user is indeed logged out.
Expected Results	● After clicking on the "Logout" button, the user should be redirected to the login page or the homepage as an unauthenticated user. ● Any user-specific information or session data should be cleared upon logout. ● Attempting to access restricted pages or functionalities after logout should result in redirection to the login page or an appropriate error message indicating that authentication is required.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 17	Test Type/Category	Integration Testing	Severity	Critical
Objective	Verify interview process has started				
Test Steps	● Log in to the application as a candidate. ● View current interviews displayed in the candidate dashboard. ● Click on the "Take Interview" button to initiate the interview process.				
Expected Results	● The user should be redirected to the interview recording page to begin the interview process.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 18	Test Type/Category	Integration Testing	Severity	Critical
Objective	Verify interview process has finished				
Test Steps	● Log in to the application as a candidate. ● View current interviews displayed in the candidate dashboard.				

	● Click on the "Take Interview" button to initiate the interview process. ● Record the interview video for a while. ● If time remains, click on the "Finish Interview" button. ● Wait to be redirected to the dashboard.
Expected Results	● The interview should be added to the list of past interviews on the user's dashboard. ● The status of the interview should be updated to "In Analysis."
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 19	Test Type/Category	Integration Testing	Severity	Critical
Objective	Verify call recording can be stored				
Test Steps	● Log in to the application as candidate. ● View current interviews displayed in candidate dashboard ● Click on take interview button. ● Record interview video for a while. ● Click on finish interview button if time candidate has remaining time ● Wait for being redirected to dashboard				
Expected Results	● Recruiter should be able to display the video of the candidate without any errors. ● Machine learning analysis of the interview should commence without errors.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 20	Test Type/Category	Integration Testing	Severity	Major
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Objective	Verify a conversation can be started
Test Steps	● Login to the application. ● Navigate to the Chat feature within the application. ● Initiate a conversation with another user.
Expected Results	● A conversation window should open, allowing communication with the selected user.
Date Tested and Test Result	08.05.2024 - NA

Test ID	Test 21	Test Type/Category	Integration Testing	Severity	Critical
Objective	Verify candidate can take the interview				
Test Steps	● Log in to the application as a candidate. ● Access the interview section of the application. ● Select an available interview to take.				
Expected Results	● The candidate should be able to start and complete the interview without errors.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 22	Test Type/Category	Integration Testing	Severity	Major
Objective	Verify recruiter can view candidates on current interviews				
Test Steps	● Log in to the application as a recruiter. ● Navigate to the current interviews section.				
Expected Results	● Recruiters should be able to see a list of candidates currently in interviews.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 23	Test Type/Category	Integration Testing	Severity	Critical
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Objective	Verify candidate can view interview status
Test Steps	- Log in to the application as a candidate. - Navigate to the interview section.
Expected Results	Candidates should be able to view the status of their ongoing interviews.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 24	Test Type/Category	Integration Testing	Severity	Critical
Objective	Verify candidate can view past interviews				
Test Steps	- Login to the application as a candidate. - Access the past interviews section of the candidate dashboard. - Select a past interview to view details.				
Expected Results	Candidates should be able to access and view details of their past interviews.				
Date Tested and Test Result	08.05.2024 - NA				

Test ID	Test 25	Test Type/Category	Integration Testing	Severity	Critical
Objective	Verify logout				
Test Steps	- Login to the application. - Click the Logout button from any page within the application.				
Expected Results	- A confirmation message should be displayed indicating successful logout. - The user should be directed to the Login page.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 26	Category	Integration	Severity	Critical
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Objective	Verify Recording Start
Test Steps	● Login to the system as a candidate. ● View current interviews on the dashboard. ● Click on the Start Interview button.
Expected Results	● The recording indicator should show that recording has started. ● User can continue recording his/her interview.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 27	Category	Integration	Severity	Critical
Objective	Verify Recording Stop				
Test Steps	● Login to the system as a candidate. ● View current interviews on the dashboard. ● Click on the Start Interview button. ● After recording for the desired amount of time (under the limitations), click on Stop Recording.				
Expected Results	● The recording indicator should stop, and the recorded file should be saved without errors. ● The user should be notified stating that the video is saved. ● The user should be directed to the dashboard.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 28	Category	Component	Severity	Major
Objective	Test Cases for Password Field on Change Password Page				
Test Steps	● Verify if the password field is accessible by the user.				

	● Ensure that the user can input text into the password field. ● Validate if the password field accepts numeric characters. ● Confirm if the password field enforces the length requirement of at least 8 characters, including both numbers and letters, and disallows special characters.
Expected Results	● The password field should be accessible and editable by the user. ● The user should be able to type in the password field without any issues. ● Numeric characters should be accepted as valid input in the password field. ● The password field should enforce the specified length requirement of at least 8 characters, including both numbers and letters, and disallow any special characters.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 29	Category	Component	Severity	Critical
Objective	Verify that Interview Guide Form is filled				
Test Steps	● Open the Interview Guide Form within the application. ● Check if all mandatory fields in the form are accessible and editable. ● Enter valid data into each mandatory field, including keywords, questions, and answers. ● Optionally, fill out any additional fields or sections provided in the form.				
Expected Results	● All mandatory fields in the Interview Guide Form should be accessible and editable. ● Valid data should be successfully entered into each mandatory field without encountering any errors. ● Optionally filled fields or sections should also accept valid data without errors.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 30	Category	Integration	Severity	Major
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Objective	Verify recruiter can create the interview guide
Test Steps	● Log in to the application as the recruiter. ● Navigate to the interview guides page. ● Click on the "Create New Guide" or similar button to initiate the creation process. ● Fill out the required fields in the interview guide form, including keywords, questions, and answers. ● Optionally, add additional details or sections to the interview guide as needed. ● Click on the "Submit" or "Create" button to save the newly created interview guide. ● Verify if the guide creation process completes without errors. ● Optionally, navigate back to the interview guides page to confirm the presence of the newly created guide.
Expected Results	● The recruiter should be able to access the interview guide creation options without encountering any errors. ● The interview guide creation form should allow the recruiter to input necessary details such as keywords, questions, and answers. ● After submission, the system should process the request without errors, and the newly created interview guide should be saved successfully. ● If any errors occur during the creation process (e.g., validation errors, server errors), appropriate error messages should be displayed to guide the user.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 31	Category	Integration	Severity	Major
Objective	Verify recruiter can edit the interview guide				
Test Steps	● Log in to the application as the recruiter. ● Navigate to the interview guides page. ● Select one interview guide to be modified. ● Click on the Edit button. ● Modify one or more fields in the interview guide form (e.g., keywords, question, answer). ● Click on the "Submit" button. ● Verify if the changes are saved successfully. ● Optionally, verify if the updated interview guide information reflects correctly on the recruiter's interview guides page.				
Expected Results	● The recruiter should be able to access their interview guide				

	editing options without encountering any errors. • The interview guide editing form should allow the recruiter to modify the keywords, questions, answers etc. • After submitting the changes, the system should process the request without errors, and the changes should be saved successfully. • If any errors occur during the editing process (e.g., validation errors, server errors), appropriate error messages should be displayed to guide the user.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 32	Category	Integration	Severity	Critical
Objective	Verify recruiter can delete the interview guide				
Test Steps	• Log in to the system with credentials as a recruiter. • Navigate to the interview guide section. • Click the Delete button for the desired interview guide.				
Expected Results	• The specific interview guide should be deleted without errors.				
Date Tested and Test Result	08.05.2024 - Pass				

5.2. Non-Functional Test Cases

Test ID	Test 33	Category	Usability	Severity	Critical
Objective	Verify the bias elimination and user-friendliness of the application interface.				
Test Steps	• Navigate through the application interface on both platforms (web and mobile). • Check for language, imagery, and design that might introduce bias. • Assess the ease of use in navigating through different functionalities. • Evaluate the comprehensibility of the reports provided to recruiters. • Interact with the pre-elimination system to understand its manageability.				
Expected Results	• The application interface is intuitive and free from bias-inducing elements.				

	• Navigating through the application is straightforward and user-friendly. • Reports provided to recruiters are easily understandable and comprehensive.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 34	Category	Reliability	Severity	Major
Objective	verify the availability and accuracy of the system under varying conditions.				
Test Steps	• Monitor the application's uptime over a period of 24 hours. • Simulate heavy load conditions to assess system response. • Test the system's accuracy edge with candidates having various accents or dialects. • Validate the system's resilience to minimal downtime during maintenance and updates. • .				
Expected Results	• The application is available 24/7 with minimal downtime for maintenance. • System maintains an accuracy level of at least 90% under normal conditions. • Performance remains consistent even during heavy load conditions.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 35	Category	Performance	Severity	Critical
Objective	Verify evaluate the performance of the application in handling high candidate volumes and report generation.				
Test Steps	• Simulate 500 candidates taking interviews simultaneously. • Measure the time taken to analyze and summarize each interview report. • Evaluate the response times for different functionalities of the application. • Monitor data extraction and machine learning analysis efficiency.				

Expected Results	• The application supports 500 simultaneous interviews without performance degradation. • Each interview report is prepared within 10 minutes. • Response times for various actions within the application meet performance expectations.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 36	Category	Supportability	Severity	Major
Objective	Verify the compatibility of the application with different operating systems and feedback mechanism.				
Test Steps	• Access the application from Windows, MacOS, Linux, Android, and iOS devices. • Verify that the application functions properly on each operating system. • Provide feedback through the designated mechanism and assess the responsiveness of the system to incorporate user inputs.				
Expected Results	• The application is accessible and functional across all specified operating systems. • User feedback is acknowledged and utilized for updating the machine learning model.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 37	Category	Scalability	Severity	Critical
Objective	Ensure the application's readiness for increased user demand and traffic peaks.				
Test Steps	• Review the roadmap for enhancements and scalability plans. • Simulate increased user demand to assess system response. • Validate the system's ability to handle fluctuations in the job market and recruitment seasons.				
Expected Results	• The scalability roadmap is comprehensive and addresses				

	potential user increases. • The system remains stable and responsive during high-demand periods.
Date Tested and Test Result	08.05.2024 - Pass

Test ID	Test 38	Category	Security	Severity	Critical
Objective	Verify User is Logged Out of the System Correctly				
Test Steps	• Check if the user's account is still available after log out. • Check if the user's information is cached after log out.				
Expected Results	• The user is logged out of the system after logout button is clicked. • The user cannot access to his/her account without login if the user is already logged out.				
Date Tested and Test Result	08.05.2024 - Pass				

Test ID	Test 39	Category	Maintenance	Severity	Major
Objective	Verify Reliability of Analysis Results				
Test Steps	• Apply software updates or patches to the system. • Verify that new features or bug fixes do not introduce regressions. • Test backward compatibility with older versions of the system. • Monitor system performance and stability after updates. • Validate data integrity and consistency post-update.				
Expected Results	• Updates and patches should be applied smoothly without causing disruptions. • New features or bug fixes should not adversely affect existing functionality. • Data integrity and consistency should be maintained across updates.				
Date Tested and	08.05.2024 - Pass				

Test Result	
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6. Maintenance Plan and Details

To ensure JobTalk remains efficient and effective, a comprehensive maintenance plan is essential. This plan includes regular updates to address bugs, security vulnerabilities, and compatibility issues, ensuring the system's integrity and security. Performance optimizations will also be a focus, enhancing the system's ability to handle increased loads, especially during peak usage times. Additionally, new features will be introduced based on user feedback and market needs, such as improvements to the NLP algorithms for refined data analysis.

The plan includes rigorous database management, with regular backups to prevent data loss and updates to database schemas as necessary when new functionalities are added or existing ones are changed. Infrastructure will be continuously monitored, with real-time system health checks and automated alerts for potential issues. Traffic management strategies like load balancing and auto-scaling will be implemented to ensure seamless user experiences.

A dedicated support team will be available to address user issues and queries promptly. This team will also collect and analyze user feedback, which is crucial for the iterative improvement of JobTalk. Regular testing routines, including performance, integration, and security tests, will be established to ensure updates do not compromise the system's functionality or security. Quality metrics will be monitored to maintain high standards and user experience guidelines.

Furthermore, the system will expand its multilingual support to include major global languages, enhancing accessibility for non-English speaking users. Continuous improvements in UI/UX will also be made to ensure the platform is accessible to users with disabilities, adhering to global accessibility standards.

These efforts aim to provide a reliable, secure, and user-friendly platform that evolves according to user needs and technological advancements in the recruitment industry. By adhering to this detailed maintenance plan, JobTalk will continue to offer high-quality services that meet the dynamic demands of its users.

7. Other Project Elements

7.1. Consideration of Various Factors in Engineering Design

7.1.1. Considerations

7.1.1.1. Public Health Considerations

JobTalk is an application that automates the recruitment process, it has no additional public health risks for the users of the application.

7.1.1.2. Public Safety Considerations

The application does not require any external hardware from the user, therefore it does not create public or individual safety concerns.

7.1.1.3. Public Welfare Considerations

Since JobTalk aims to provide companies with a more efficient recruitment process by removing human bias and creating a faster evaluation process, it will contribute to public welfare by ensuring equal opportunities for applicants.

7.1.1.4. Global Considerations

JobTalk provides recruitment services for English-speaking candidates and employers. Since English is considered a global language nowadays, so JobTalk is an application that can be used globally. Additionally, we aim to include more languages in the future.

7.1.1.5. Cultural Considerations

Using machine learning algorithms JobTalk eliminates human bias, creating more culturally diverse working environments for companies.

7.1.1.6. Social Considerations

In the app, some personal information such as phone number, email address, and interview footage is stored. However, to protect the privacy of the user, this data cannot be accessed by any parties not involved in the recruitment process for the specific job that the candidate has applied for.

7.1.1.7. Environmental Considerations

JobTalk is an app that automates the recruitment process and it does not have any direct or indirect effect on the environment to be considered.

7.1.1.8. Economic Considerations

The application is expected to be financed by yearly subscription fees paid by the companies. Candidates applying for jobs do not pay any fees. JobTalk will be funded by the subscription fees.

Topic	Effect level	Effect
Public health	0	No effect.
Public Safety	0	No effect.
Public Welfare	9	Recruitment process will be handled more efficiently and faster by JobTalk, contributing to public welfare.
Global	8	JobTalk's evaluations are done in English.
Cultural	8	JobTalk will contribute to a socially diverse working environment by eliminating human bias.
Social	4	JobTalk requires some personal data but manages the access of the data.
Environmental	0	No effect.
Economic	6	Subscription fee is paid by the companies.

7.1.2. Constraints

7.1.2.1. Data Privacy Constraints

The protection of personal information such as phone number, address, and birthday. The protection of this data is crucial for personal privacy reasons. Additionally, the interviews will be recorded on the server and stored for 6 months.

7.1.2.2. Functionality Constraints

With JobTalk, we aim to revolutionize the recruitment processes by creating a new way for pre-interview processes. Therefore, the functionality of the system is one of our main focuses during the design process. We aim to ease the authentication and testing process by generating a report on a single video interview consisting of multiple questions, taking into account:

- General content summary,
- Pauses (i.e umm),
- Emotional state

by using the voice data from the interview recording performed by the candidate.

7.1.2.3. Cost Constraints

The application will require a server machine with high computational power for interview recording and interview evaluation with machine learning models and a Raspberry Pi 4 that runs the frontend (React application). There are also Collab virtual machines that are used to train the models with hourly fees. In addition, GPT models used have costs per token. For now, the cost of these machines is compensated by the team, later to be compensated through yearly subscription fees.

7.1.3. Standards

The standards used for this project are as follows:

- IEEE 830 for software requirements specification
- ISO/IEC 29148 for requirements engineering
- UML 2.5.1 in Enterprise Architect for UML diagrams (User Stories, Use Cases)
- Requirement Traceability Matrix (RTM)
- Version Control (Git and Github in collaboration with JIRA)

7.2. Ethics and Professional Responsibilities

In developing JobTalk, we prioritize ethical standards and professional responsibilities. Key to our approach is ensuring the security and privacy of all data collected, adhering to regulations like GDPR to prevent unauthorized access and breaches. We are particularly careful about the use of cameras during interviews; all candidates are asked for explicit consent before any recording starts, ensuring they are fully aware and agreeable to the use of their visual data.

At JobTalk, addressing the potential for bias in AI-driven recruitment is a top priority. We strive to eliminate the traditional biases found in recruitment by using algorithms that objectively analyze candidate data. Transparency is also central to our ethical approach; we provide detailed documentation on how our algorithms make decisions, allowing users to fully understand and challenge these decisions if needed.

7.3. Teamwork Details

7.3.1. Contributing and functioning effectively on the team

One of the most important aspects of the project is teamwork. We separate the workload as equally as possible, taking into account every member's strong suits. The project has many parts that require interest and expertise. We divide tasks into machine learning, backend, frontend and deployment categories to distribute the work as efficiently as possible. When it comes to the reports we try to separate the work according to each member's responsibilities in the implementation process.

During the implementation process, backend development and database integration is performed by Elifsena Öz and Tuğrul Sağlam. System design, administration and frontend development is done by Barış Yıldırım. İpek Öztaş and Emre Uğur are responsible for machine learning research and model implementation. We plan two meetings each week, one to talk to our supervisor and one to come together as a team to work. The first type of meeting is usually at most 30 minutes long discussing the previous week's progress and the task distribution for the next week. The second type of meeting generally takes around 4-5 hours, working together and helping each other where needed.

7.3.2. Helping creating a collaborative and inclusive environment

As a group we communicate easily both through synchronous and asynchronous channels. For setting up meetings and sharing news on the project we use our Whatsapp group. We do not have strict day and times for our meetings, we try to find the best slot for all members. Our meetings with our supervisor are held via Zoom and our weekly work meetings are done face-to-face in school. Compared to our online meeting schedule last semester, we have found that getting together as a team for 4-5 hours a week helps us work together more efficiently and is better for making important decisions.

7.3.3. Taking lead role and sharing leadership on the team

In our team we do not have an assigned leader. Each of our members have equal say and have the opportunity to lead when needed. Leadership means taking responsibility for the whole team and managing the process in the best way possible. We believe that each member should feel these responsibilities and act accordingly. Even though we do not have a leader for the whole project, categories and specific tasks from the project are distributed among our team members to lead.

- Research and ML Model Development is led by Emre Uğur
- Testing and Bugfix is led by İpek Öztaş
- Deployment, Hosting and Frontend Development is led by Barış Yıldırım
- Core Backend Development and Documentation is led by Tuğrul Sağlam
- Backend Integration of Submodules and Database Implementation is led by Elifsena Öz

7.3.4. Meeting objectives

Initially, our goal was to develop a cross-platform application to streamline the recruitment process by incorporating video recording, video processing, and generating analysis reports. However, we soon recognized that both candidates and recruiters predominantly conduct their interviews on PCs. Consequently, we shifted our focus towards developing a web application, concentrating on enhancing our features. Originally, we intended to use the NextCloud environment for video recording, but due to multiple access issues, we opted to develop our own recorder. This adjustment resolved many challenges and significantly improved the development of our application. These strategic decisions were made during

group meetings and under the guidance of our advisor. For significant decisions, we ensured consensus among all group members and obtained approval from our advisor as well.

7.4. New Knowledge Acquired and Applied

Throughout the development of JobTalk, our team gained substantial insights and expertise in several key areas of technology and project management. This section highlights the new knowledge we acquired and how we applied it effectively.

Implementing AI and machine learning models was an important part of our project. We learned to train, test, and refine various models to analyze interview data accurately. The process involved understanding the intricacies of algorithm optimization, data preprocessing, and feature selection to enhance the performance of our models. This not only improved our technical proficiency but also deepened our understanding of how machine learning can be applied to solve real-world problems in recruitment.

Our project benefited greatly from adopting agile project management methodologies. Through regular sprints and iterative development, we learned to manage tasks efficiently, adapt to changing requirements, and deliver results on time. Effective communication and collaboration tools were integral to coordinating our efforts, particularly in a multi-disciplinary team that included members with varying technical backgrounds.

Deploying and hosting a scalable web application was a significant learning curve for the team. We gained practical experience in cloud services and server management, which involved setting up secure and robust environments for our application to run smoothly. Understanding the deployment pipelines and continuous integration/continuous deployment (CI/CD) practices were crucial skills that we developed during this phase.

Quality assurance was critical in ensuring the reliability and usability of JobTalk. We learned advanced testing techniques, including unit testing, integration testing, and performance testing. This not only helped in catching bugs early but also in maintaining the quality of the application as new features were added. The practice of systematic testing and validation reinforced the importance of building a product that meets user expectations and industry standards.

8. Conclusion and Future Work

The development of the JobTalk system marks a significant advancement in applying machine learning within the recruitment sector. This project showed the critical value of efficient project management and teamwork in addition to its technical proficiency. The system's deployment demonstrated careful planning and strong execution, highlighting the importance of hosting solutions, disciplined deployment methodologies, and a focus on testing and quality assurance. Together, these elements have produced a scalable and efficient tool that enhances the recruitment process by making it more streamlined and data-centric.

Looking to the future, the JobTalk system is set to embrace several innovative enhancements that will further transform the candidate evaluation process. Plans include the integration of language proficiency tools to more accurately assess a candidate's language skills, which are crucial for roles requiring specific linguistic abilities. Additionally, incorporating body language analysis could offer deeper insights into a candidate's non-verbal communication and confidence, providing a richer evaluation framework.

The system will also expand to support multiple languages, broadening its accessibility to a global audience and ensuring wider reach and inclusivity. Introducing elements of gamification could make the recruitment process more engaging for candidates, thereby enhancing their experience and potentially improving the quality of applications. Lastly, a commitment to continuous learning and enhancement of the AI models will ensure that the system remains at the forefront of technology, adapting to new trends and advancements in machine learning.

We can offer a custom model or fine-tuning service to companies that can provide us with human-labeled scores from interviews. Scoring 100 to 200 interviews is sufficient to develop a custom model. However, if a company cannot provide this amount, they can opt for fine-tuning instead. This approach allows us to tailor our analysis according to the company's preferences. Additionally, we can enhance our system by introducing a ranking functionality. Instead of merely assigning scores to interviews, our system can rank a set of interviews, enabling recruiters to easily identify the top performers. This feature is motivated by the observation that the Area Under the Curve (AUC) scores of our regression models are

superior to our Pearson correlation coefficients. For instance, for a particular position, a company can interview a set of candidates and our system can sort them based on their performance. While a recruiter can manually sort candidates by examining the scores, this built-in functionality saves both time and effort.

These forthcoming features will not only refine the capabilities of the JobTalk system but also ensure its continued relevance and efficacy in the rapidly evolving recruitment technology landscape. By consistently evolving and integrating new functionalities, JobTalk aims to set a new standard in recruitment processes, benefiting both employers and candidates.

9. Glossary

PostgreSQL: An object-relational database management system.

SQLAlchemy: A Python library that allows users to manage SQL databases in Python language.

ORM (Object Relational Mapping): Mapping programming code to database structures.

NAS (Network Attached Storage): Storage of data that can be accessed remotely.

SSL encryption: SSL is standard technology for securing an internet connection by encrypting data sent between a website and a browser (or between two servers)

HTTP Flood Attack: Volumetric distributed denial-of-service (DDoS) attack designed to overwhelm a targeted server with HTTP requests.

DOS (Denial of Service): A type of attack in which a malicious actor interrupts the normal functioning of a device making it unavailable.

NGINX: An open-source software used for serving web content and as a reverse proxy.

Reverse Proxy: A web service that stands in front of servers managing client requests to increase security, reliability and performance.

Brute-force Attack: A hacking technique that breaks encryption keys, passwords, and login credentials by trial and error.

XSS (cross-site scripting): An attack where the code of a reliable program or website is injected with harmful executable scripts by the attacker.

JSX: Javascript XML allows usage of HTML in React.

AUC: The AUC score is indicative of how effectively a model can order the inputs.

10. References

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